



Welcome to Fire-IQ, Lesson 1: Mechanisms of Fire

Central Fire-IQ Slide Deck

Last Updated August 31, 2026 by Chiara Pesce



A Scientific Perspective: What is Fire?

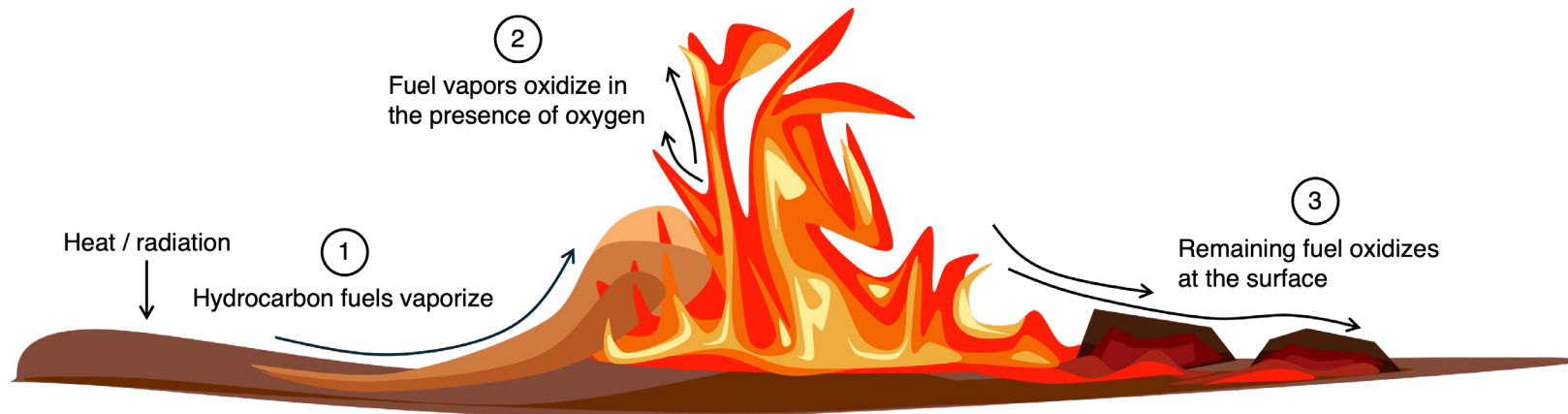


Fire as a Combustion Process

Combustion: the rapid oxidation of fuel that produces heat and gases

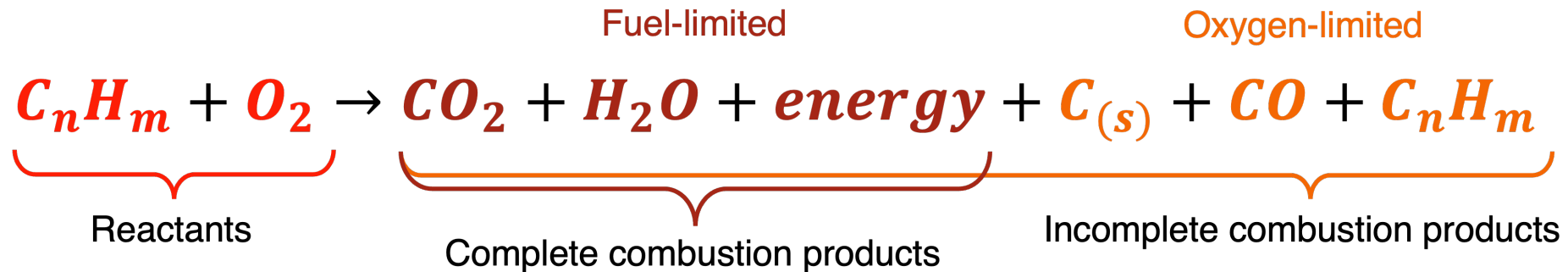
Combustion typically occurs in three steps:

1. Organic fuels are heated, vaporizing water and hydrocarbons with low boiling points (**pyrolysis**)
2. Vaporized hydrocarbons oxidize, releasing energy and combustion products (**flaming combustion**)
3. Remaining carbon and minerals oxidize (**smoldering / glowing combustion**)



Energy & Chemical Transformations

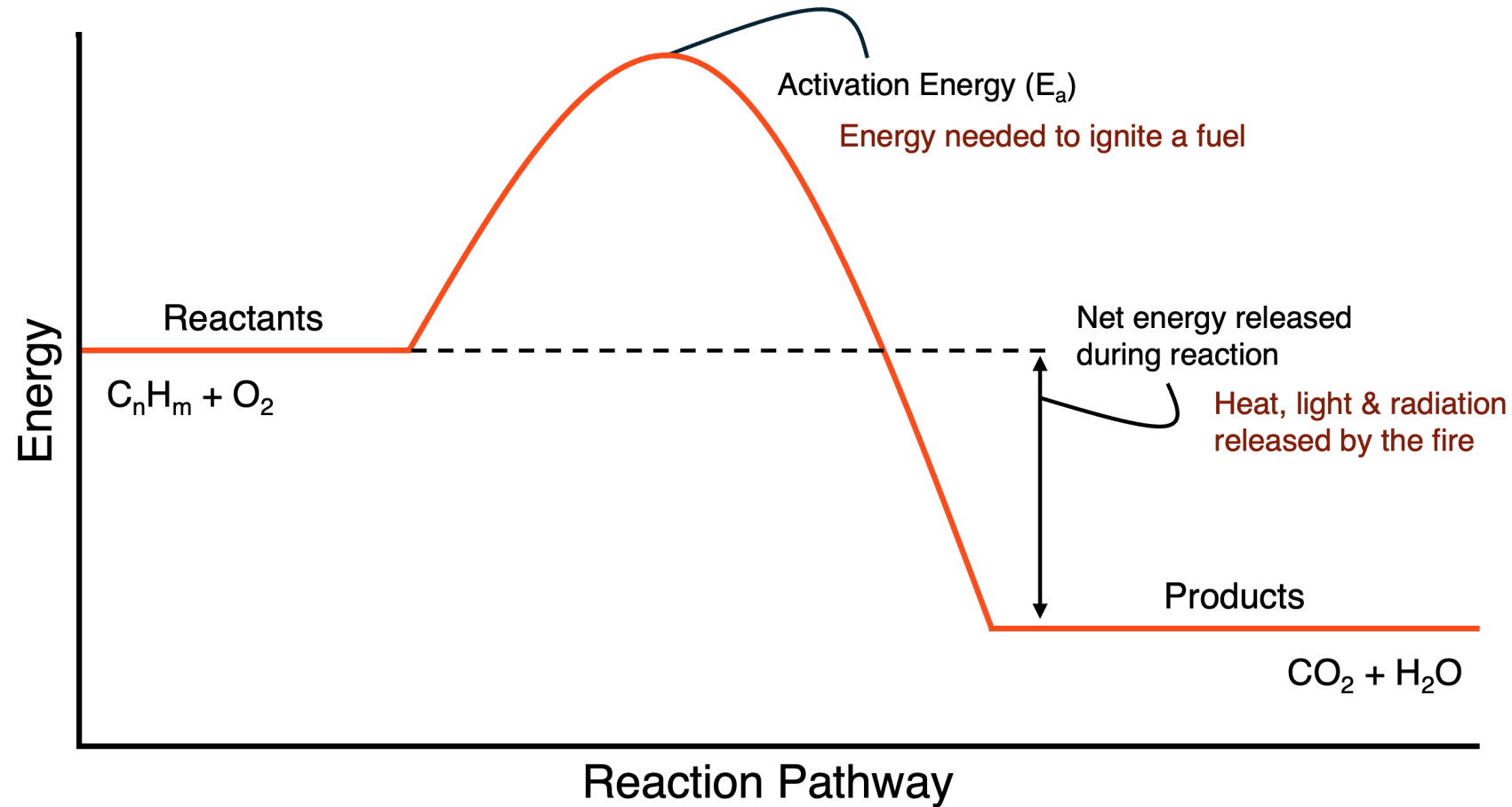
- Fire is an exothermic oxidation reaction where atoms rearrange into more stable lower-energy forms; the excess energy released as heat or radiation



- Products from fire depend on the amount of oxygen available at the combustion site, dividing into incomplete & complete combustion



Energy & Chemical Transformations



An Engineering Perspective: How is Fire Used?

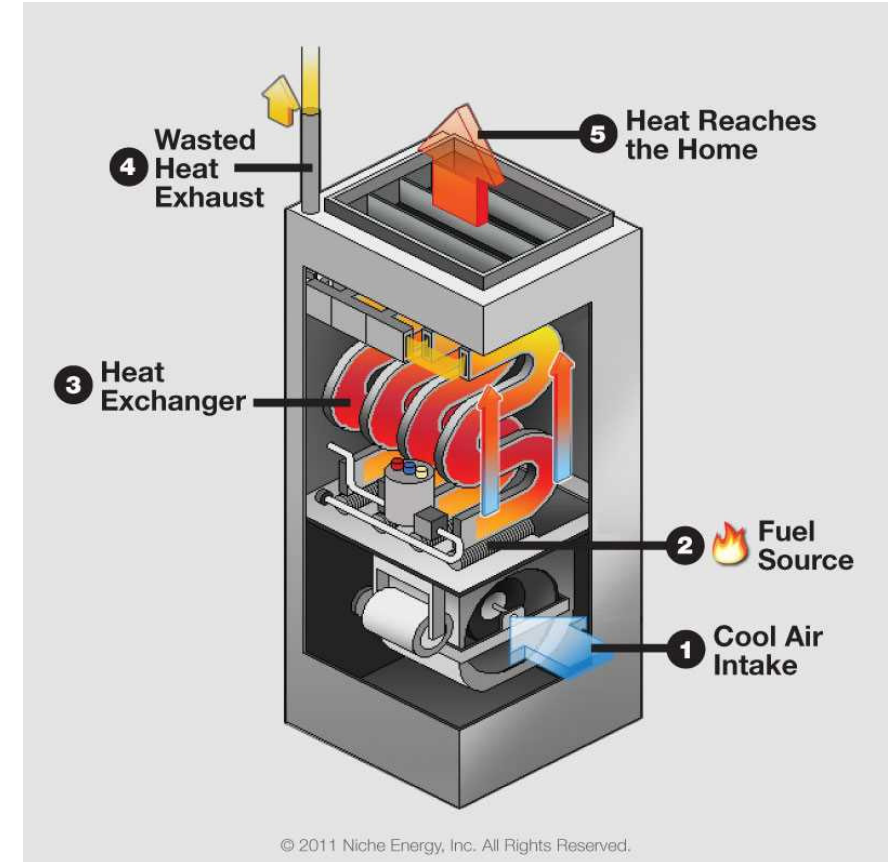


Heat & Energy from Combustion

Heat from combustion **evaporates** water, often producing high pressure steam whose **thermal energy** is converted to **mechanical work** or distributed for residential heating.



Source: <https://kianturbotec.com/gas-turbine-efficiency/>



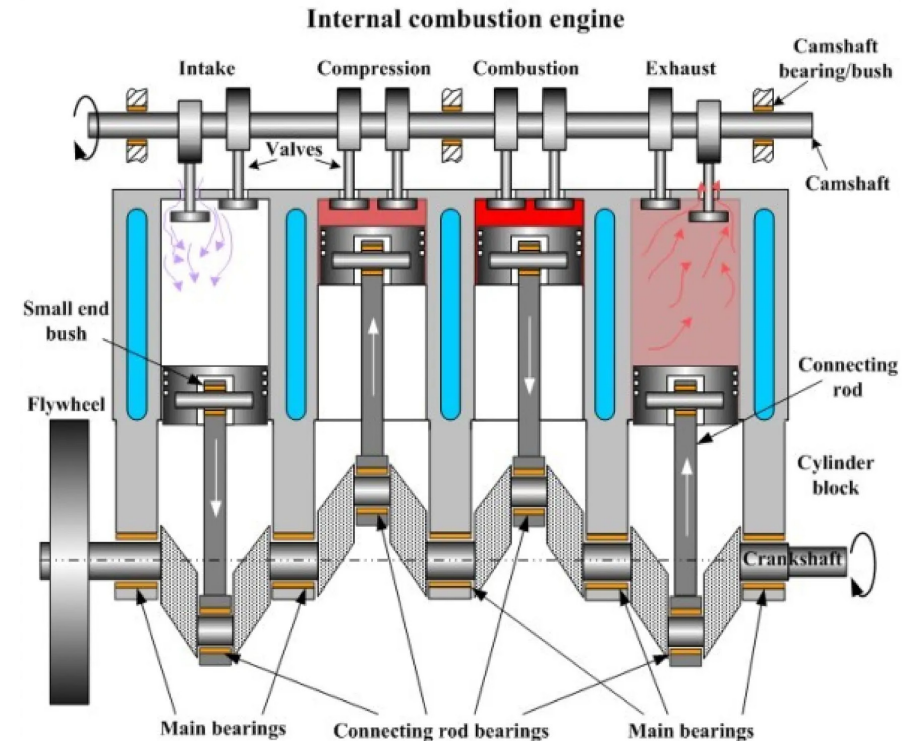
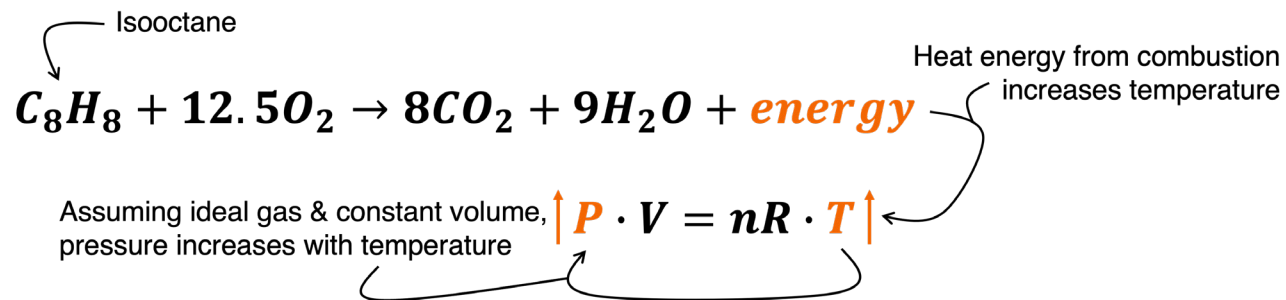
Source: <https://wiringpictures.net/carrier-gas-furnace-diagram>



Motion from Combustion

Combustion has long been harnessed for motion, whether it be steam engines or modern internal combustion engines.

In an internal combustion engine, oxidation of hydrocarbon fuel releases heat at **constant volume**, raising gas temperatures to ~2500 K while exerting pressures ~60 bar on cylinders. The pressurized gas then **expands** against the piston, converting chemical energy to thermal energy, to mechanical work.



Source: Mehran et al., "A Critical Review on Physical Vapor Deposition Coatings Applied on Different Engine Components."



Historical & Modern Uses of Fire

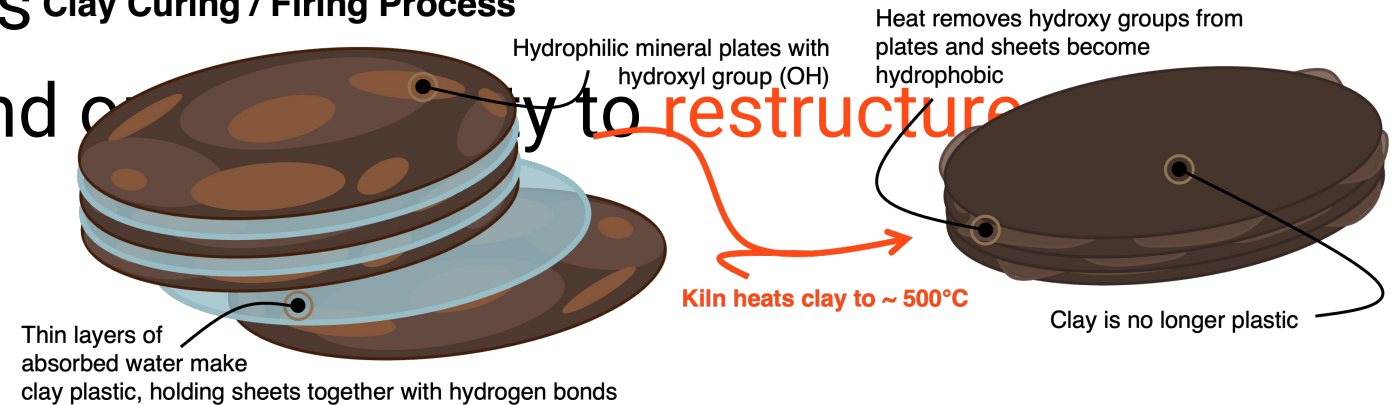
- Many indigenous peoples have used fire to **cultivate** and **steward** landscapes
- Fire use in cooking has made food more accessible & nutritious
- Combustion is used in a wide variety of industrial &



Source: dmhoang. "Rethinking Wildfire." UC Davis, October 1, 2020. <https://www.ucdavis.com/climate/news/rethinking-wildfire>.

Processes Clay Curing / Firing Process

depend on **clay to restructure**

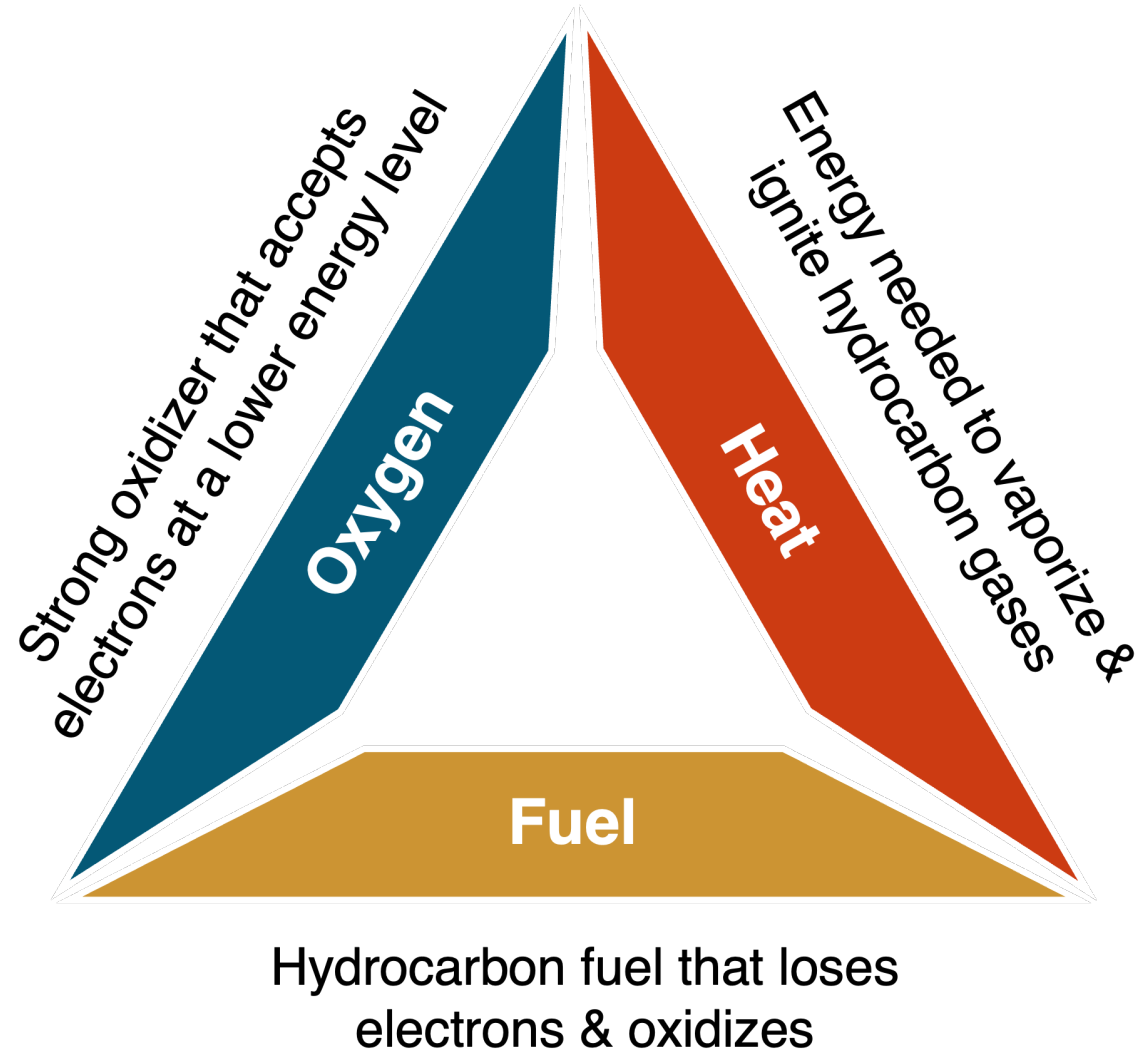


What Factors Influence Fire Behavior?



The Fire Triangle

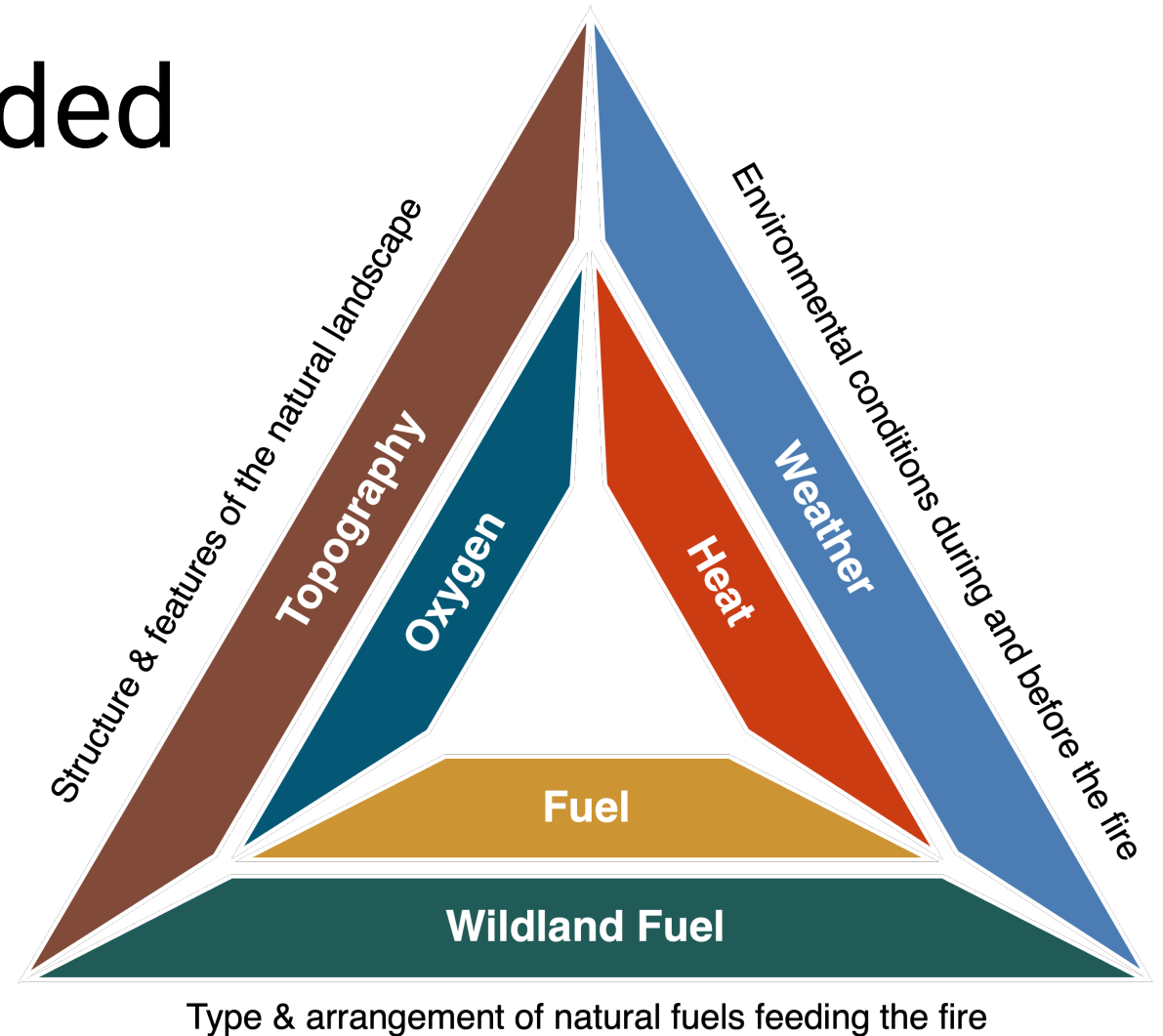
- The fire triangle depicts the key **factors** that allow a fire to ignite and burn
- If any element of the triangle is taken away, fire will no longer be able to sustain itself



Fire Triangle Expanded

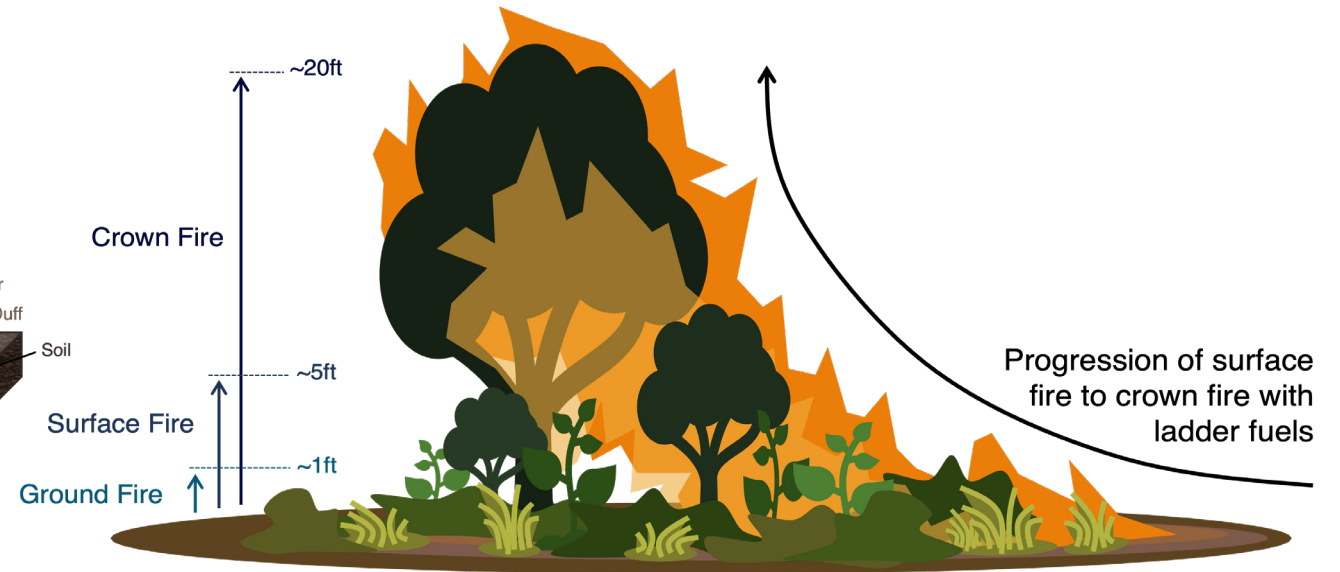
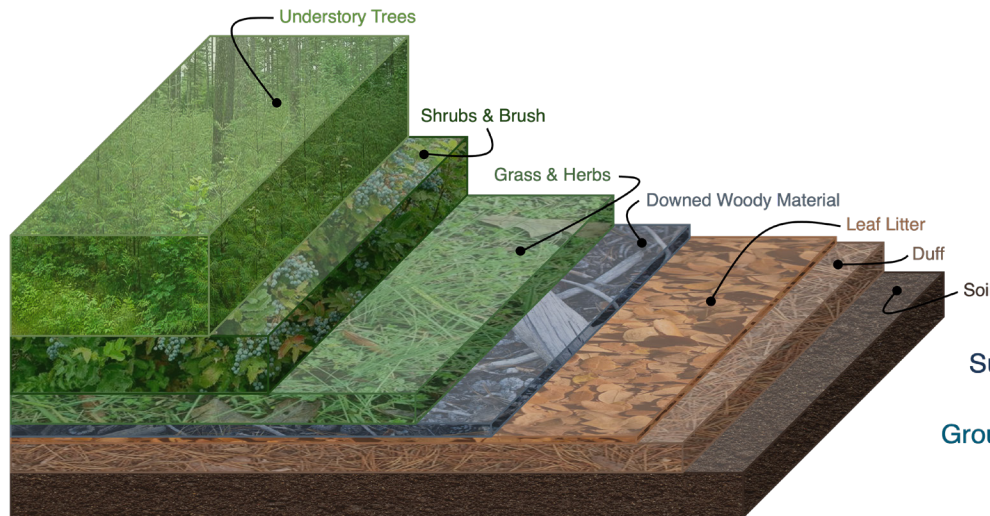
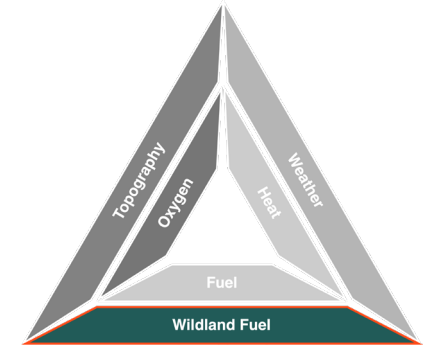
- The second tier of the fire triangle depicts factors that influence how fire behaves & spreads after ignition

Bonus: What are other ways the fire triangle can be portrayed, or how can it be further expanded?



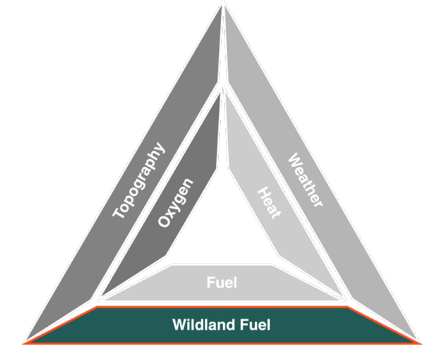
Wildland Fuel: Type & Arrangement

- Fuel types and sizes determine how quickly fuels ignite and burn
- Fuel arrangement determines how fire propagates and climbs



Wildland Fuel: Fuel Moisture

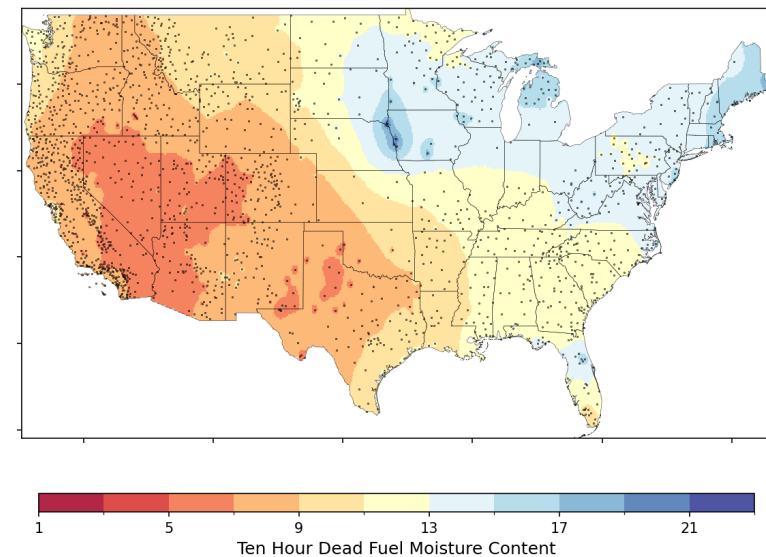
- Water's **latent heat of evaporation** increases the amount of heat needed to ignite fuel sources
- High moisture content in leaves & foliage prevents **torching** where fire consumes crown foliage
- Drought-stressed trees increase forest flammability



Source:

https://gacc.nifc.gov/swcc/dc/nmadc/predictive/fuels_fire-danger/fuels_fire-danger.htm

Minimum Ten Hour Dead Fuel Moisture Content (2026-08-18) (Forecast)



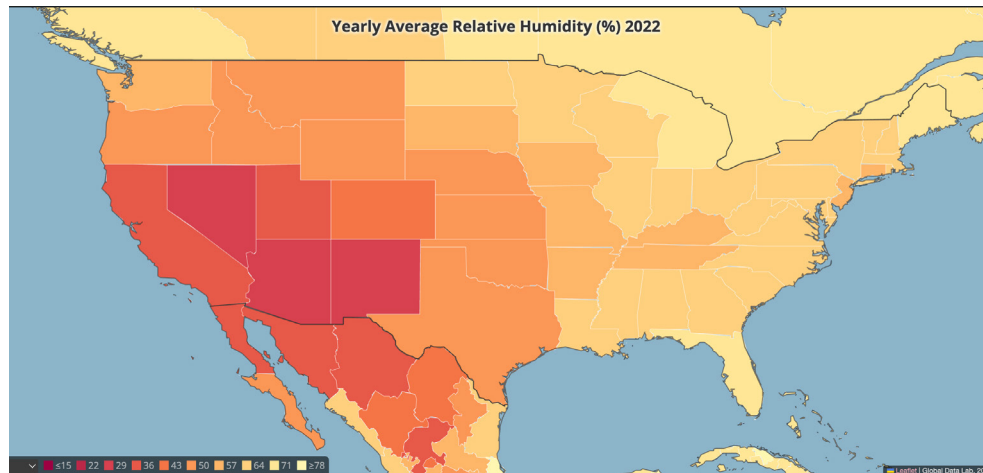
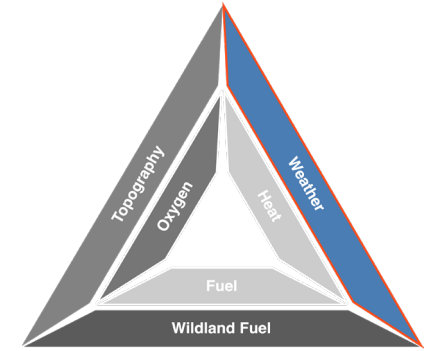
Dry-Weight Live Moisture Content (%)	Woody Fuels, Foliage, Forest Conditions
>150% to 300%	Maturing foliage still developing, fresh foliage fully expanded, vigorously growing, Early in growing cycle
>120% to 150%	Mature foliage, new growth complete and comparable to older perennial foliage. Shrubs and grasses resist spread
>100% to 120%	Mature foliage, new growth complete and comparable to older perennial foliage. Flammable shrubs should burn. Grasses resist spread.
>80% to 100%	Anticipate flammable shrubs burning aggressively
>50% to 80%	Leaves are yellowing and curing, expect shrubs to contribute to fire intensity
>30% to 50%	Dormant, leafless deciduous shrubs, increased dead fuel loads.

Source: <https://www.nwccg.gov/publications/pms437/fuel-moisture/live-fuel-moisture-content>

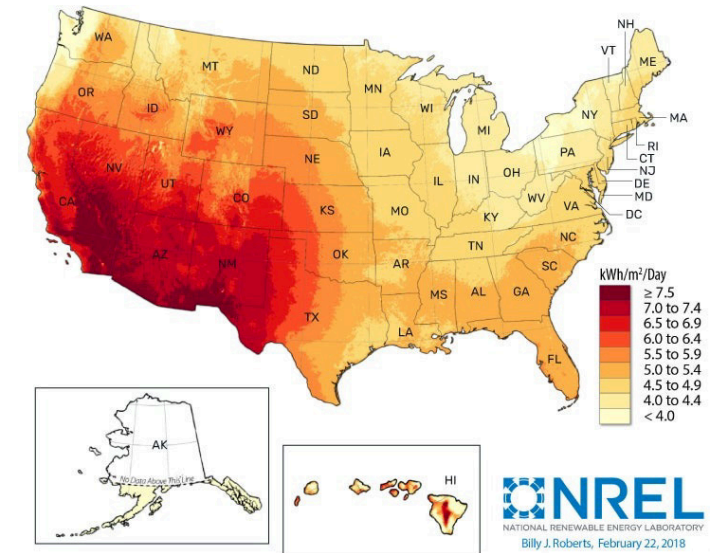


Weather: Temperature & Humidity

- High air temperatures and solar radiation raise the **surface temperature of fuels**, reducing the amount of heat needed to ignite fuel loads
- Daytime solar heating raises ambient temperatures and **reduces relative humidity**, increasing fire risk & rate of fire spread

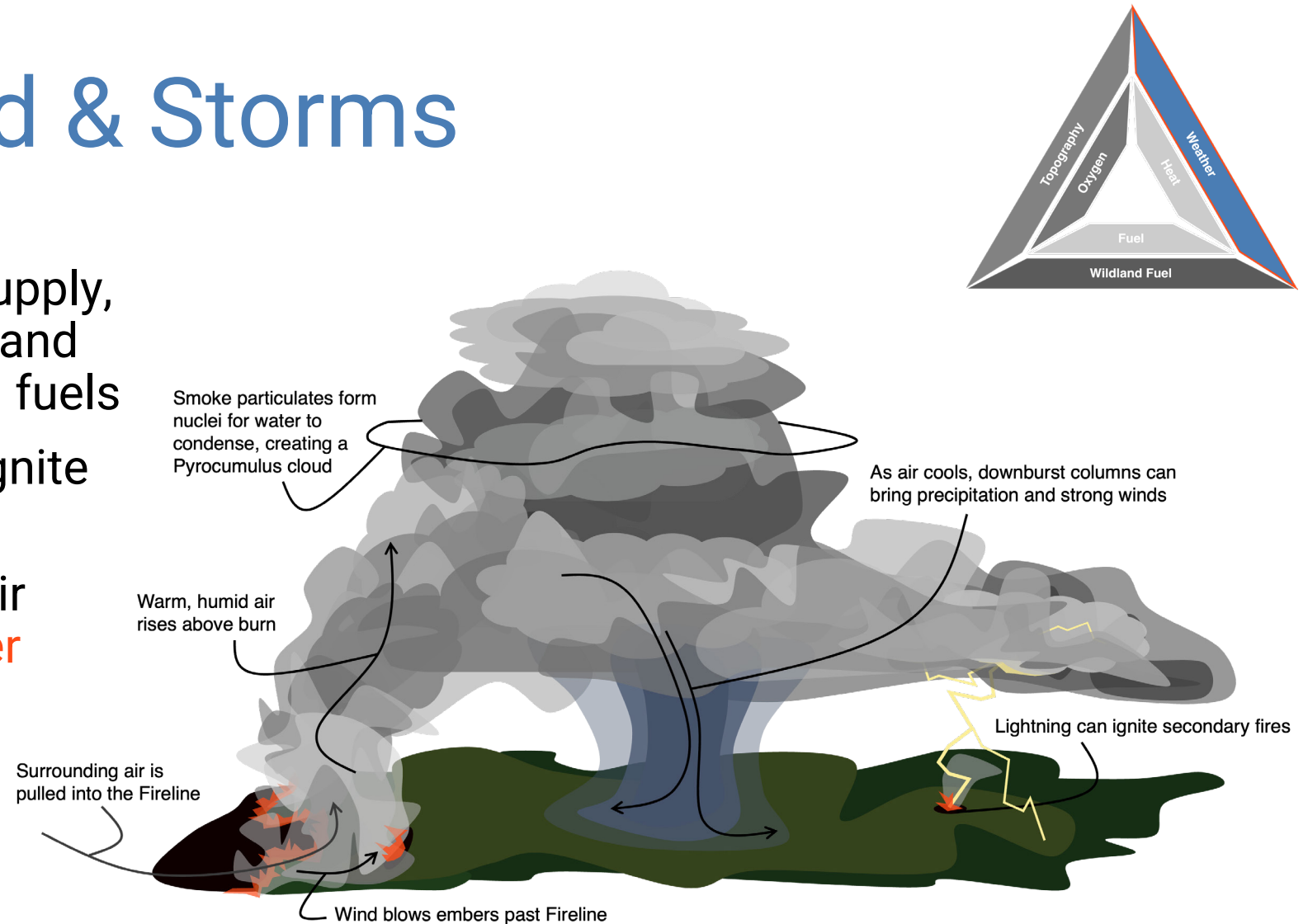


Source:
<https://globaldatalab.org/geos/maps/relhumidityyear/>



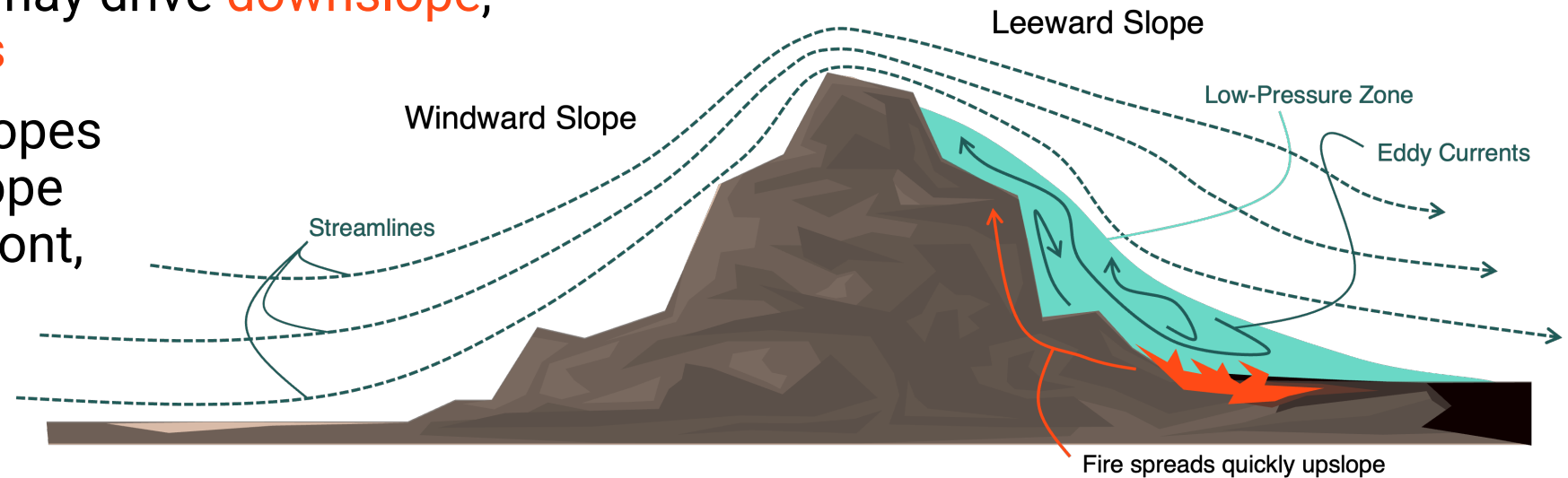
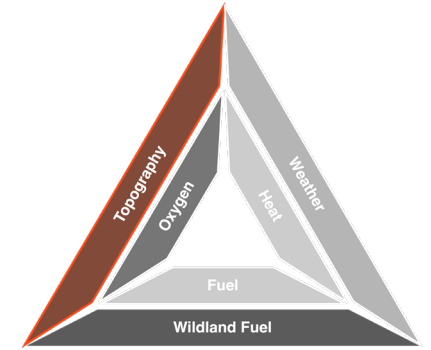
Weather: Wind & Storms

- Wind enhances oxygen supply, speeding up combustion and heat transfer to unburned fuels
- Embers carried by wind ignite **spot fires**
- Wildfires often create their own **convection** & **weather systems** that increases the rate of combustion and spread



Topography

- South-facing slopes are warm and dry whereas north-facing slopes are cool and damp, impacting vegetation & fire
- Daytime solar heating drives **upslope, upcanyon winds**
- Nighttime cooling may drive **downslope, downcanyon winds**
- Fires burning on slopes **preheat** fuels upslope from the flaming front, increasing the rate of spread

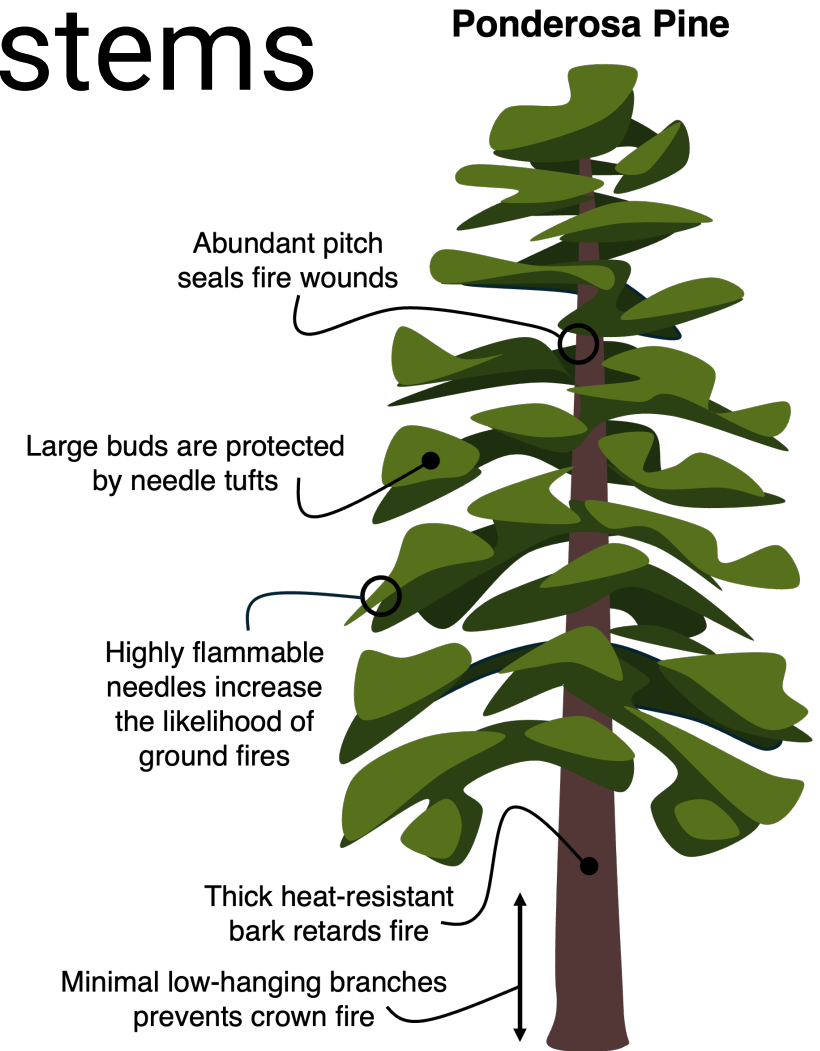


How Does Fire Impact the World around It?



Fire-Adapted Forest Ecosystems

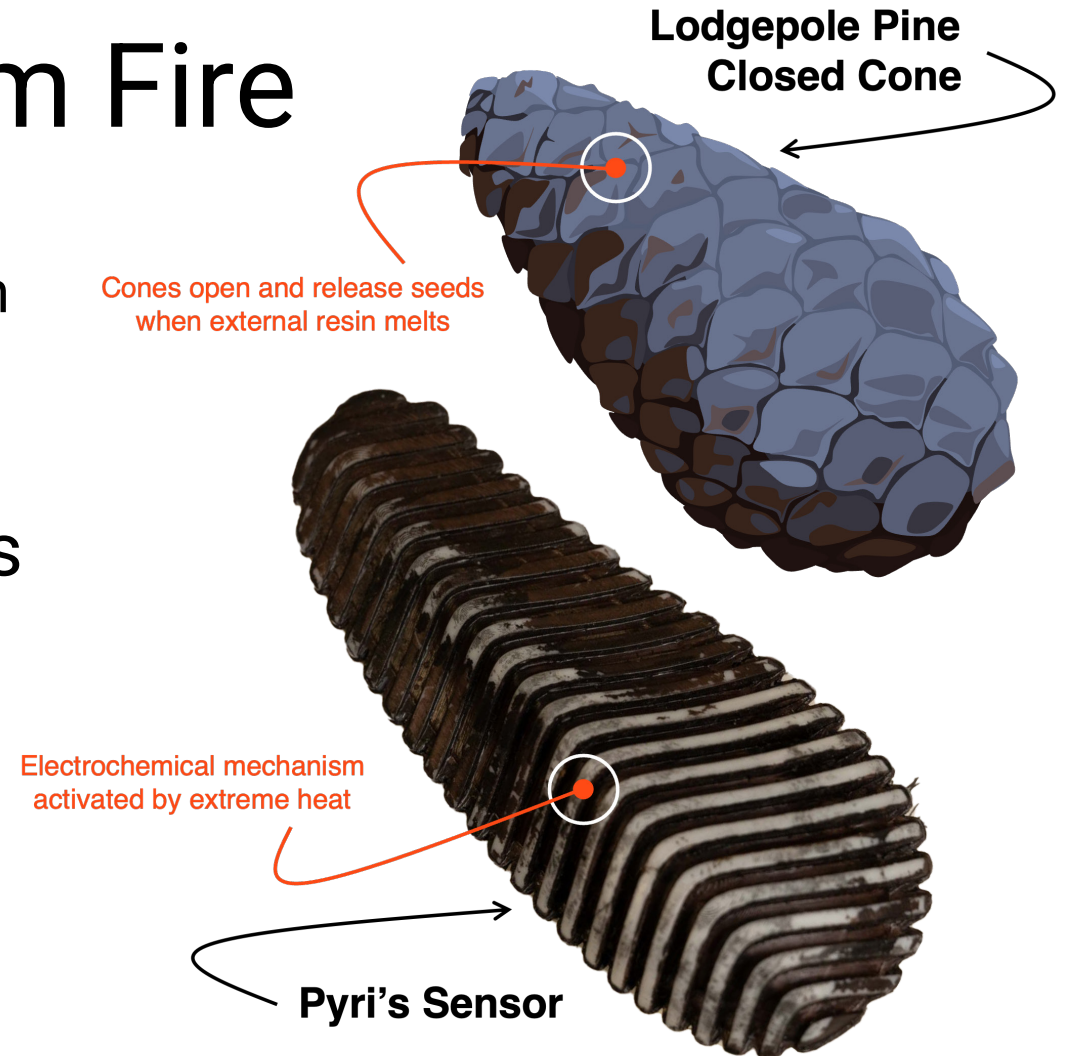
- Many trees like **Ponderosa Pine**, **Jeffrey Pine**, & **Coastal Douglas-Fir** are adapted to, or dependent on low-to-mid intensity fires
- Post fire conditions can restore nutrients to the soil, encouraging growth of grasses and native plants like **Fireweed**, **Currant** or **Clover**, nourishing wildlife such as **Elk**
- Longer, **high-intensity** fires are changing the ecology of fire-adapted forests



Solutions for Fire, from Fire

Pyri developed a low-cost wildfire detection technology based on the heat-release mechanisms serotinous cones like that of the fire-adapted **Lodgepole Pine**.

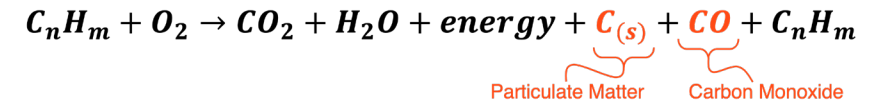
When this biodegradable sensor is exposed to heat, an internal electrochemical trigger sends a low-frequency signal to nearby authorities and communities.



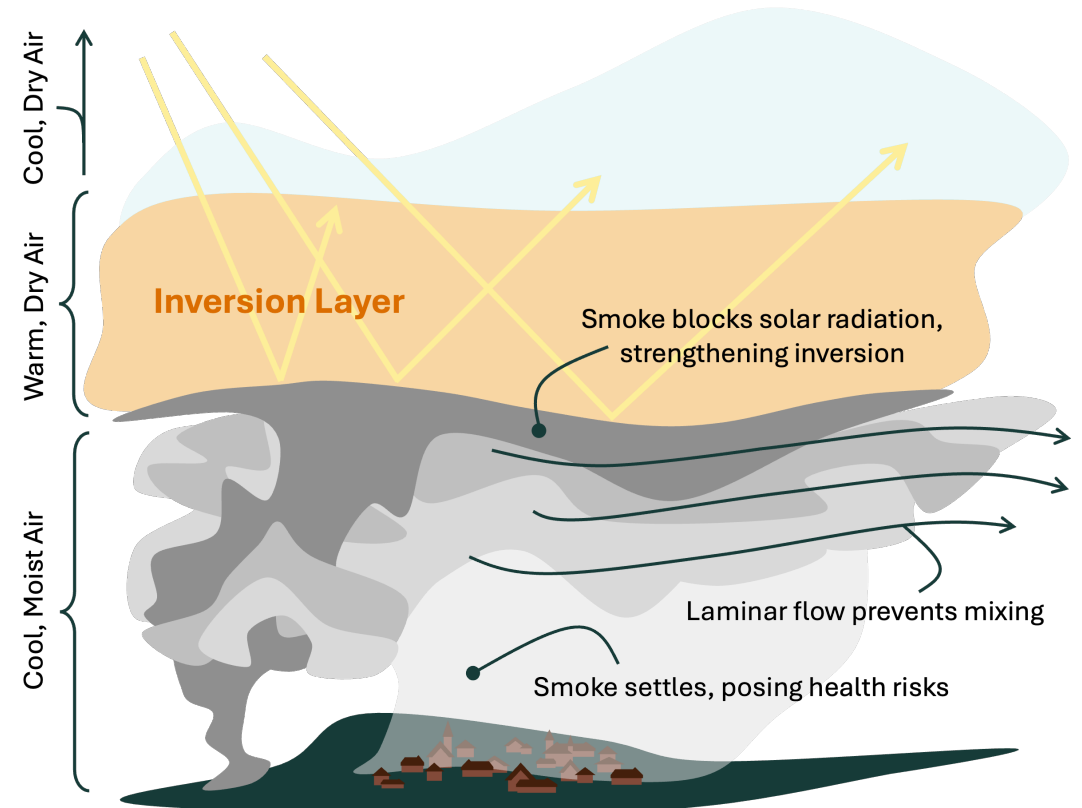
Website: <https://pyri.io/>



Air Pollution & Flooding

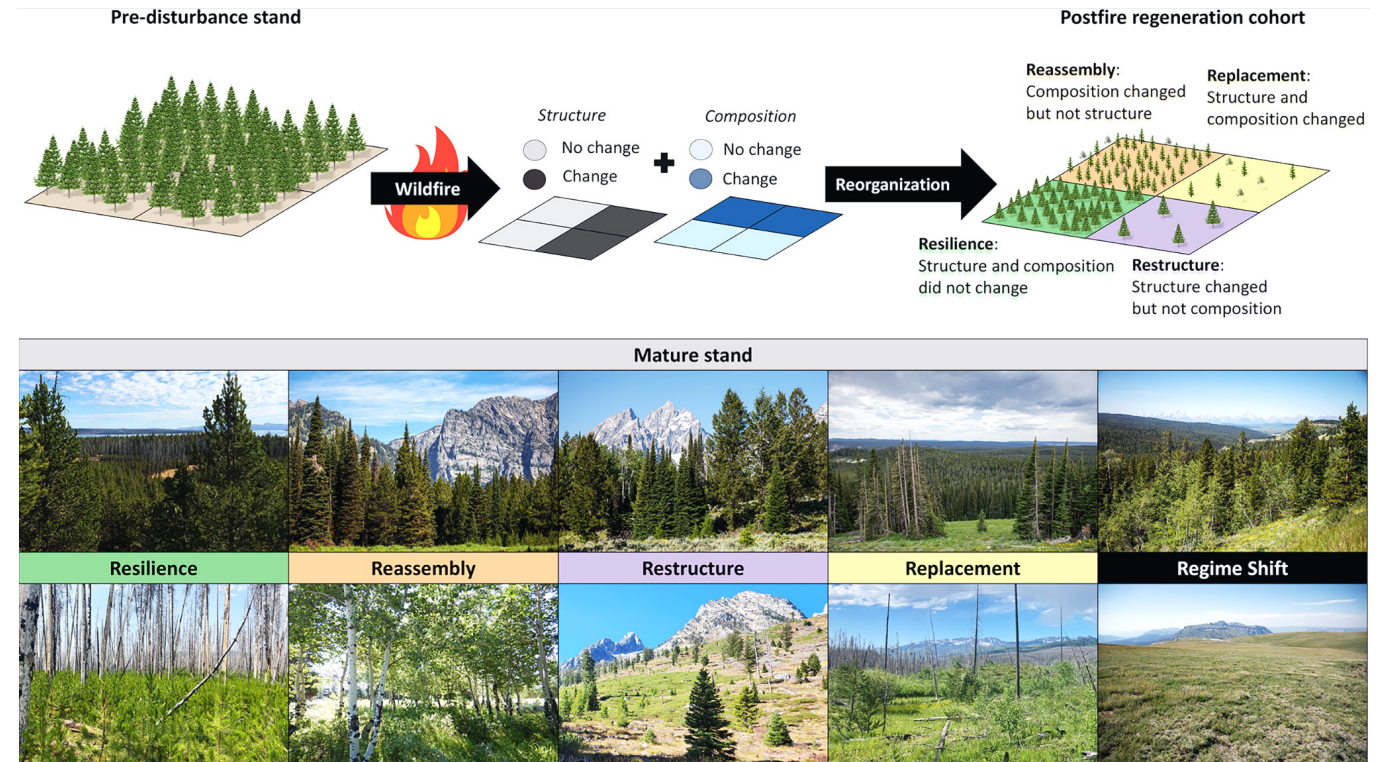


- While haze and smoke from wildfire have historically made western skies hazy, events like **inversions** and increased fire duration or intensity can be hazardous to human health
- Carbon monoxide (**CO**) and **PM2.5** (Particulate matter < 2.5 μm ϕ) produced by fire impact health
- Post burn soil often has a **water-repellent film** & lacks vegetation, making burn areas prone to **flooding** and **landslides**



Remediation & Regeneration

- Postfire ecosystem recovery depends on myriad factors including **forest composition, fire behavior, fire intensity, surrounding ecosystems** etc.
- Forest Service Burned Area Emergency Response (**BEAR**) teams are examples of human-assisted efforts to restabilize an ecosystem after wildfire



Source: Knowlton et al., "Simulated Postfire Tree Regeneration Suggests Reorganization of Greater Yellowstone Forests during the 21st Century."



Climate & Changing Conditions

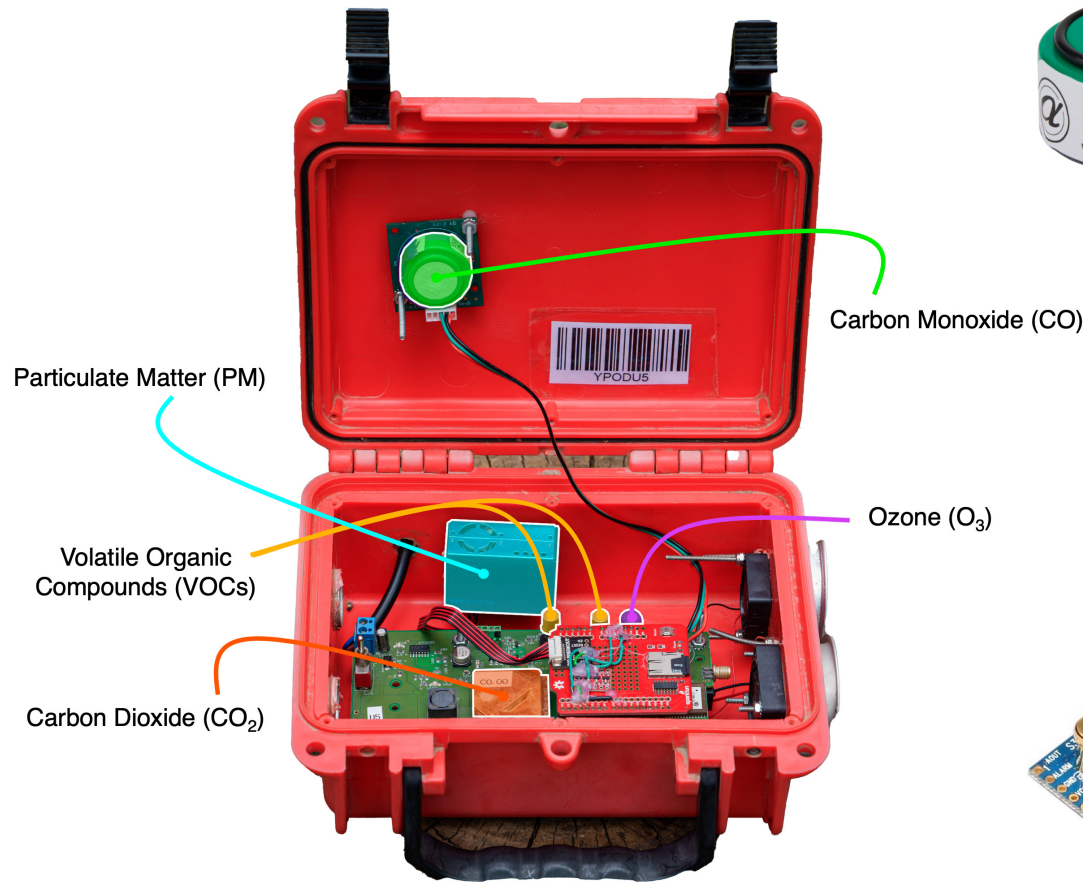
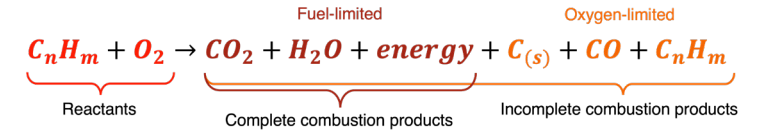
- Global warming & climate change are increasing the **frequency** and **intensity** of wildland fires in the West
- Drought, beetle / pest infestations, logging, development, and aggressive fire suppression change the **makeup** of forests and how they **respond** to fire
- Growing development on the wildland-urban interface (**WUI**) is increasing the risk wildfire poses to people
- Prescribed burning & fuel management practices are becoming more widespread in the West, with the goal of restoring a semblance of **natural disturbance processes** to the forest



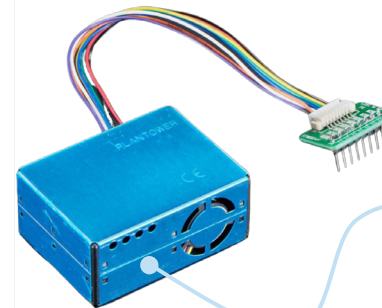
Applying Fire Mechanisms: Fuel Impact Demo



Quantifying Aspects of Fire



Carbon Monoxide (CO):
Incomplete combustion byproduct



Particulate Matter (PM):
Smoke plume & direction



Carbon Dioxide (CO₂):
Complete / flaming combustion



Quantitative vs Qualitative Observations

Quantitative Observations:

Measurable, discrete or quantifiable characteristics of a phenomena or behavior



Qualitative Observations:

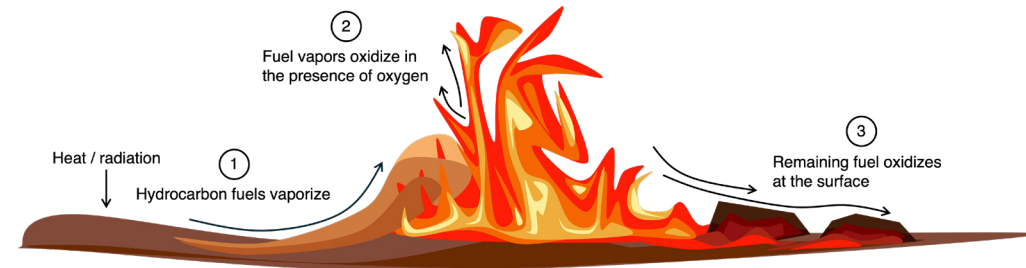
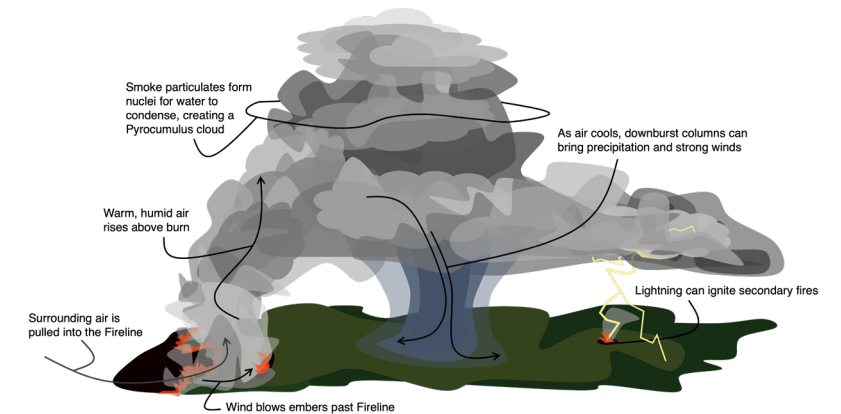
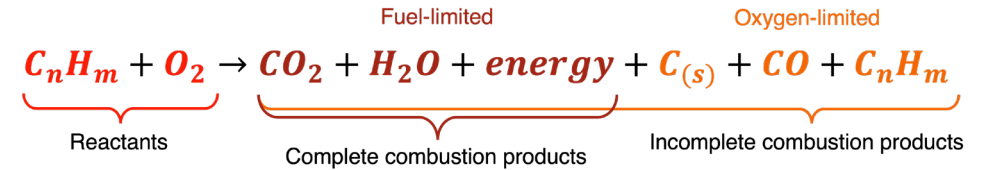
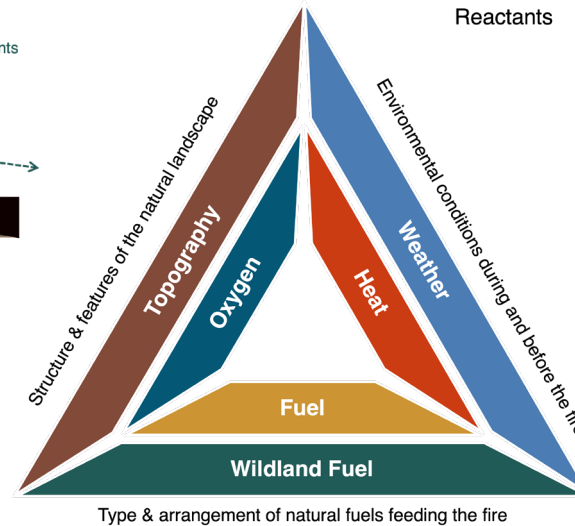
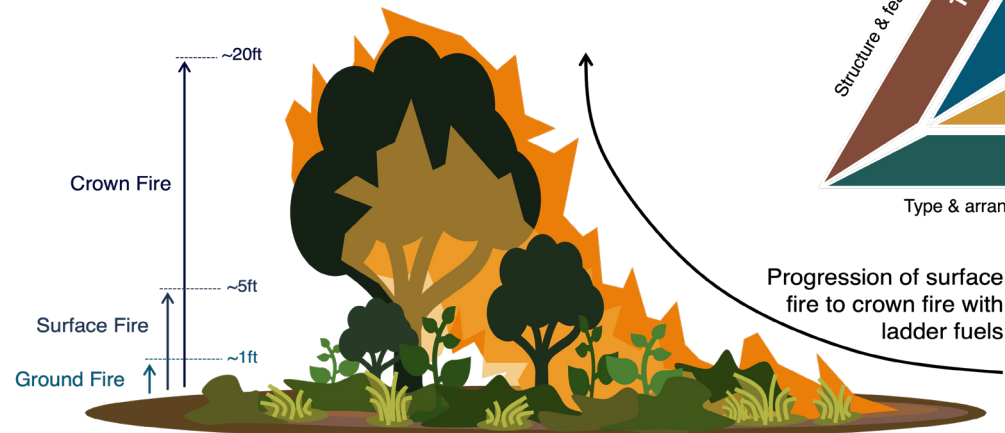
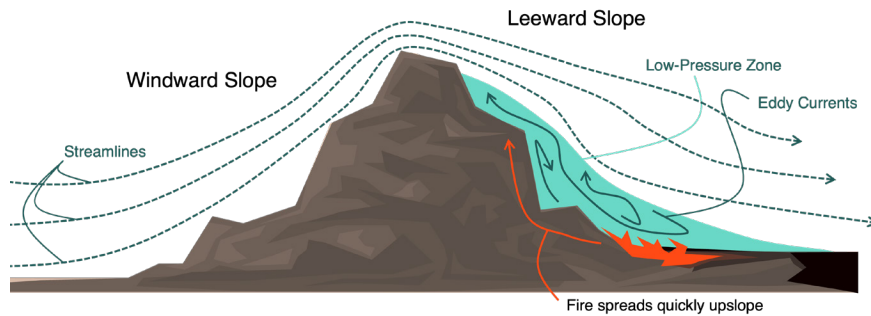
Complex, holistic, and unquantifiable characteristics of a phenomena or behavior



Bonus: what are some benefits and drawbacks of each type of observation?



Reminder: Fire Triangle



Comparison Across the Fire Triangle

The goal of this activity is to isolate a variable (ex. fuel moisture, arrangement, type, ignition, etc.) and observe how it impacts fire behavior.

Activity Steps:

1. Choose a variable to investigate (ex. **fuel moisture**, **wind speed**, **fuel arrangement**, **fuel type**, **ignition method**, etc.)
2. Build two identical fuel stacks that differ only in the chosen variable
3. Place your fuel stacks in the provided stoves and ignite!
4. Observe how the two fires differ in intensity, flame height, color, spread, temperature, etc.



If you want to learn more, we recommend starting with these excellent sources!

Finney, M. A., Cohen, J. D., Forthofer, J. M., McAllister, S. S., Gollner, M. J., Gorham, D. J., Saito, K., Akafuah, N. K., Adam, B. A., & English, J. D. (2015). Role of buoyant flame dynamics in wildfire spread. *Proceedings of the National Academy of Sciences*, 112(32), 9833–9838. <https://doi.org/10.1073/pnas.1504498112>

Giap, S. G. E., Ahmad, M. F., & Nawi, Z. M. (2025). Soil Factors Influencing Wildfire Behavior and Severity: A Review. *Environment and Ecology Research*, 13(6), 880–890. <https://doi.org/10.13189/eer.2025.130610>

Kerlin, K. (2020, October 1). Rethinking Wildfire. *UC Davis, Science & Climate*.
<https://www.ucdavis.com/climate/news/rethinking-wildfire>

NWCG Incident Response Pocket Guide (IRPG) (PMS 461, NFES 001077). (2025). National Wildfire Coordinating Group. <https://fs-prod-nwcg.s3.us-gov-west-1.amazonaws.com/s3fs-public/publication/pms461.pdf>

Smith, H. Y., & Arno, S. F. (1999). Eighty-eight years of change in a managed ponderosa pine forest. *Gen. Tech. Rep. RMRS-GTR-23*. Ogden, UT: U.S. Department of Agriculture, Rocky Mountain Research Station. 55 p., 23.
<https://doi.org/10.2737/RMRS-GTR-23>

Zhou, J., He, S., Wu, S., Zhang, J., Wang, Q., & Wang, F. (2026). THU-Wildfire: A multitemporal, multimodal observation dataset for wildfire behavior dynamics. *Environmental Modelling & Software*, 198, 106872.
<https://doi.org/10.1016/j.envsoft.2026.106872>

